

CHAPTER 3

Software

Requirement Analysis

3.3 需求获取的内容

- (1) 功能
- (2) 性能
- (3) 环境
- (4) 界面
- (5) 接口
- (6) 用户或人的因素
- (7) 文档
- (8) 数据
- (9) 资源
- (10) 安全保密
- (11) 软件成本消耗与开发进度
- (12) 质量保证

3.4 Requirement Elicitation (需求获取)

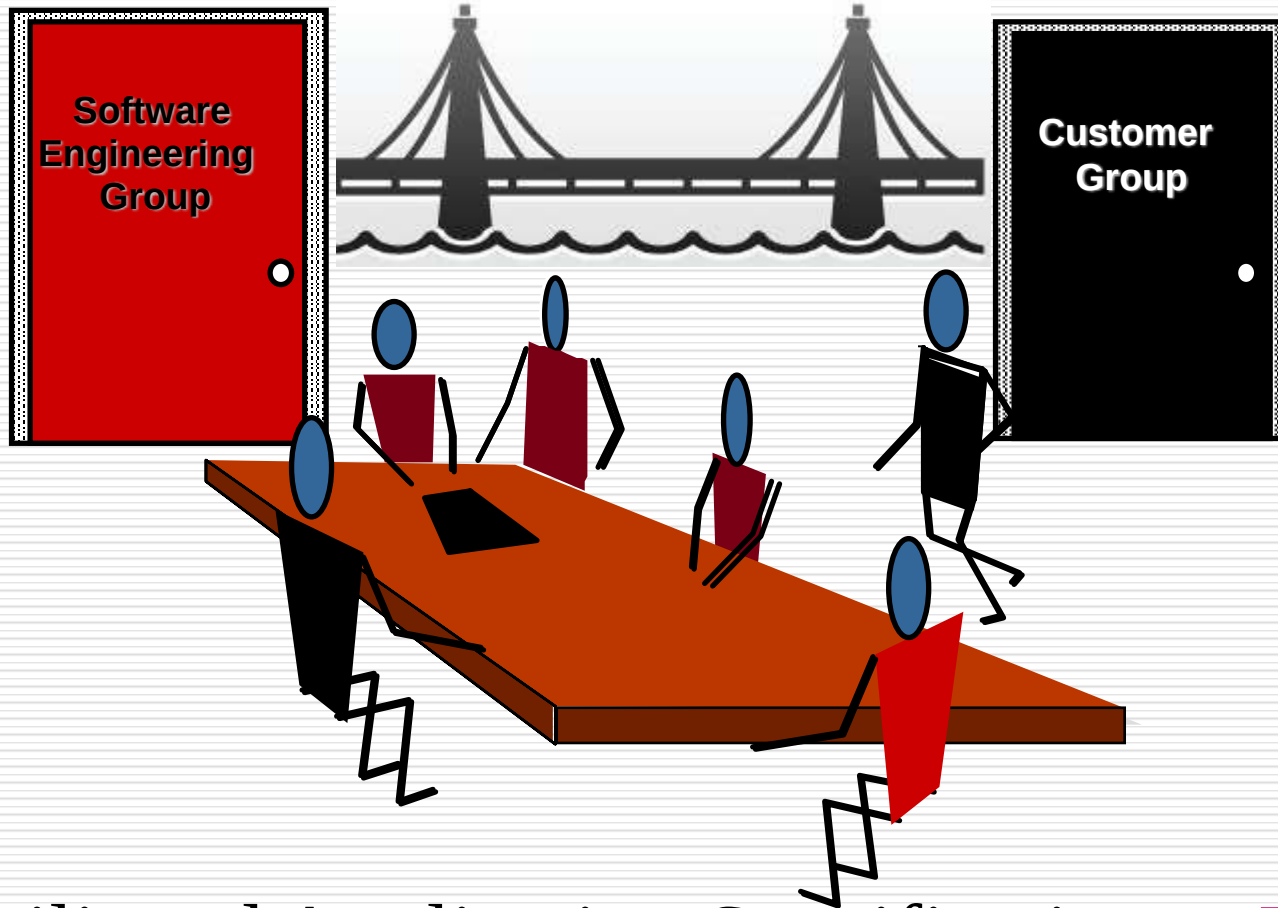
The process of discovering the client's requirement.

**Called requirement elicitation, or
requirement discovering, or
requirement collection, or
requirement capture, or
requirement gather**

Requirements Elicitation (简介)

- **任务：** 主动与干系人协同工作，找出他们的需求，识别潜在的冲突，磋商解决矛盾，定义系统范围与边界。
- **关键：** 确保需求、该问题的解决是有商业价值的
- **注意以下问题：**
 - ✓ 产品设计目标不明确
 - ✓ 干系人参与不足
 - ✓ 干系人之间缺少共识
 - ✓ 画蛇添足
 - ✓ 需求快速变化
 - ✓ 变更管理不足
 - ✓ 需求分析不足

Requirements Gathering



Facilitated Application Specification, Techniques

需求获取

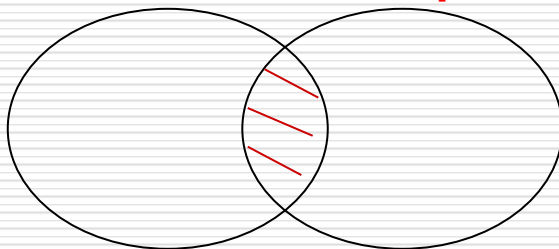
□ Stakeholders

□ 例子：HIS-健康信息系统

in individual or group requirements sessions to define system details

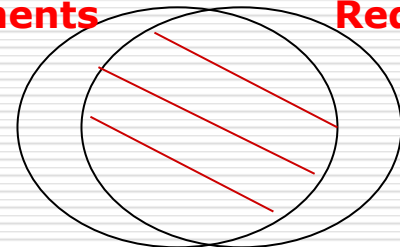
- ✓ Results in a high quality system from the customer/user point of view
- ✓ Results in higher end-user satisfaction
- ✓ Helps to educate customer/user about system through their participation

Raw Requirements **Real Requirements**



Without customer-oriented requirements-gathering practices

Raw Requirements **Real Requirements**



With customer-oriented requirements-gathering practices

需求抽取过程中，最困难的不是记录用户需求，而是与用户探讨磋商，发现真正要解决的问题，确定适用的方案。

软件工程专业， 不能只有笔记本

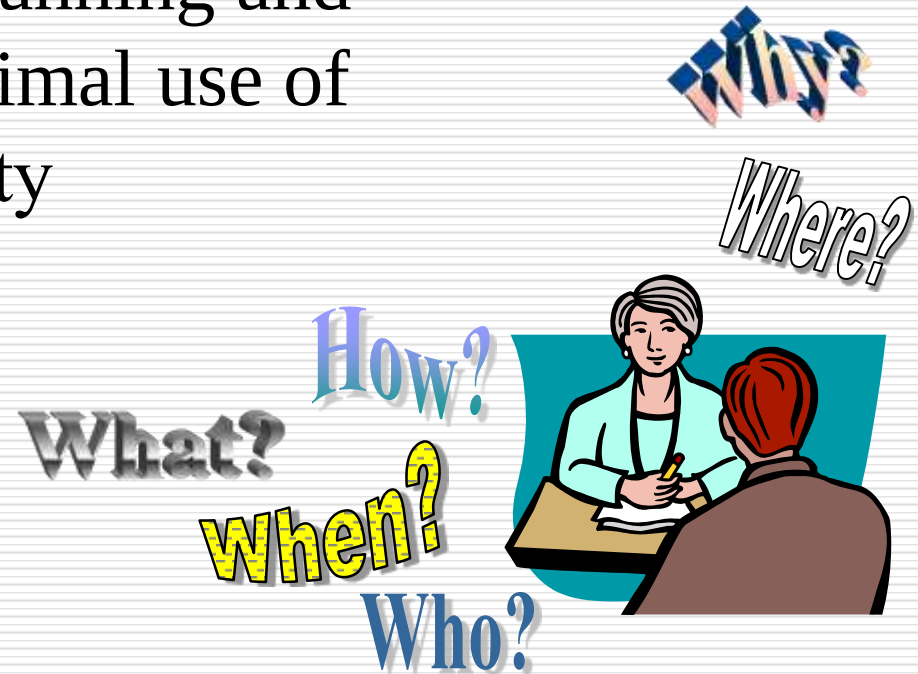


Requirements Gathering(方法)

- Interview
- Scenarios
- Viewpoints
- Videotape camera
- Inquire table
- Web searching
- Requirements community in Internet like open source
- 闭门造车—格力电器—董明珠

(1) Interview 访谈, 会谈

- Question and answer meetings in one-on-one or small group (focus group) setting
- May require detailed planning and facilitation to make optimal use of stakeholder's availability



Interview (最一般)

There are two types of interview

1. **Closed interviews** where a pre-defined set of questions are answered.
2. **Open interviews** where there is no pre-defined agenda and a range of issues are explored with stakeholders.

Interviews in practice

- ❑ Normally a mix of closed and open-ended interviewing.
- ❑ Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system.
- ❑ Interviews are not good for understanding domain requirements
 - Requirements engineers cannot understand specific domain terminology;
 - Some domain knowledge is so familiar that people find it hard to articulate or think that it isn't worth articulating.

Effective interviewers

- ❑ Interviewers should be open-minded, willing to listen to stakeholders and should not have pre-conceived ideas about the requirements.
- ❑ They should prompt the interviewee with a question or a proposal and should not simply expect them to respond to a question such as ‘what do you want’.

Before interview

➤ **Plan and schedule for interview**

- ✓ **Define topic**
- ✓ **Organize your questions.**
- ✓ **Designate roles to team members.**
- ✓ **Inform the client.**
- ✓ **Indicate the topic.**

Before interview

➤ Prepare for interview

- ✓ learn about the client's business area.
- ✓ learn about the client's organization.
- ✓ memorize client's names.
- ✓ be familiar with application domain
- ✓ be familiar with right terminology
- ✓ build a glossary
- ✓ open questions in freedom, why? when? where?
- ✓ preplanned question----form, check list

During interview

- **Open interview**
 - ✓ Introduce team.
 - ✓ Summarize previous meetings findings (if applicable).
 - ✓ Introduce topics.
- **Body of interview**
 - ✓ Make clients feel they are participants in the solution.
 - ✓ Take notes, but listen.
 - ✓ Keep it short.
 - ✓ Keep it focused.
- **Close interview**
 - ✓ Summarize.
 - ✓ Thanks clients for their time.

After interview

- ✓ **Immediately organize your notes**
- ✓ **Summarize findings**
- ✓ **Identify points still unclear**
(starting point for next)
- ✓ **Evaluate your performance**
- ✓ **.....**

(2) Scenarios (讲故事, 形象, 生动)

- ✓ story, film, real-life **rather than abstraction**
- ✓ a storyboard, a series of diagrams depicting the sequence of actions or events, a **paper prototype**
- ✓ that is, a series of sheets of paper depict screen and user's response
- ✓ Eg. ATM card, 支付宝
- ✓ use case in object oriented analysis

Scenarios

- ✓ Scenarios are descriptions of how a system is used **in practice**
- ✓ They are helpful in requirements elicitation as people can relate to these more readily than abstract statement of what they require from a system
- ✓ Scenarios are particularly useful for adding **detail** to an outline requirements description

(3) Viewpoints (观点, 视角, 干系人)

- Viewpoints are a way of structuring the requirements to represent the perspectives of different stakeholders. Stakeholders may be classified under different viewpoints.**
- This multi-perspective analysis is important as there is no single correct way to analyse system requirements.**

Types of viewpoint

➤ **Interactor viewpoints**

- ✓ People or other systems that interact directly with the system. In an ATM, the customer's and the account database.

➤ **Indirect viewpoints**

- ✓ Stakeholders who do not use the system themselves but who influence the requirements. In an ATM, management and security staff are indirect viewpoints.

➤ **Domain viewpoints**

- ✓ Domain characteristics and constraints that influence the requirements. In an ATM, an example would be standards for inter-bank communications.

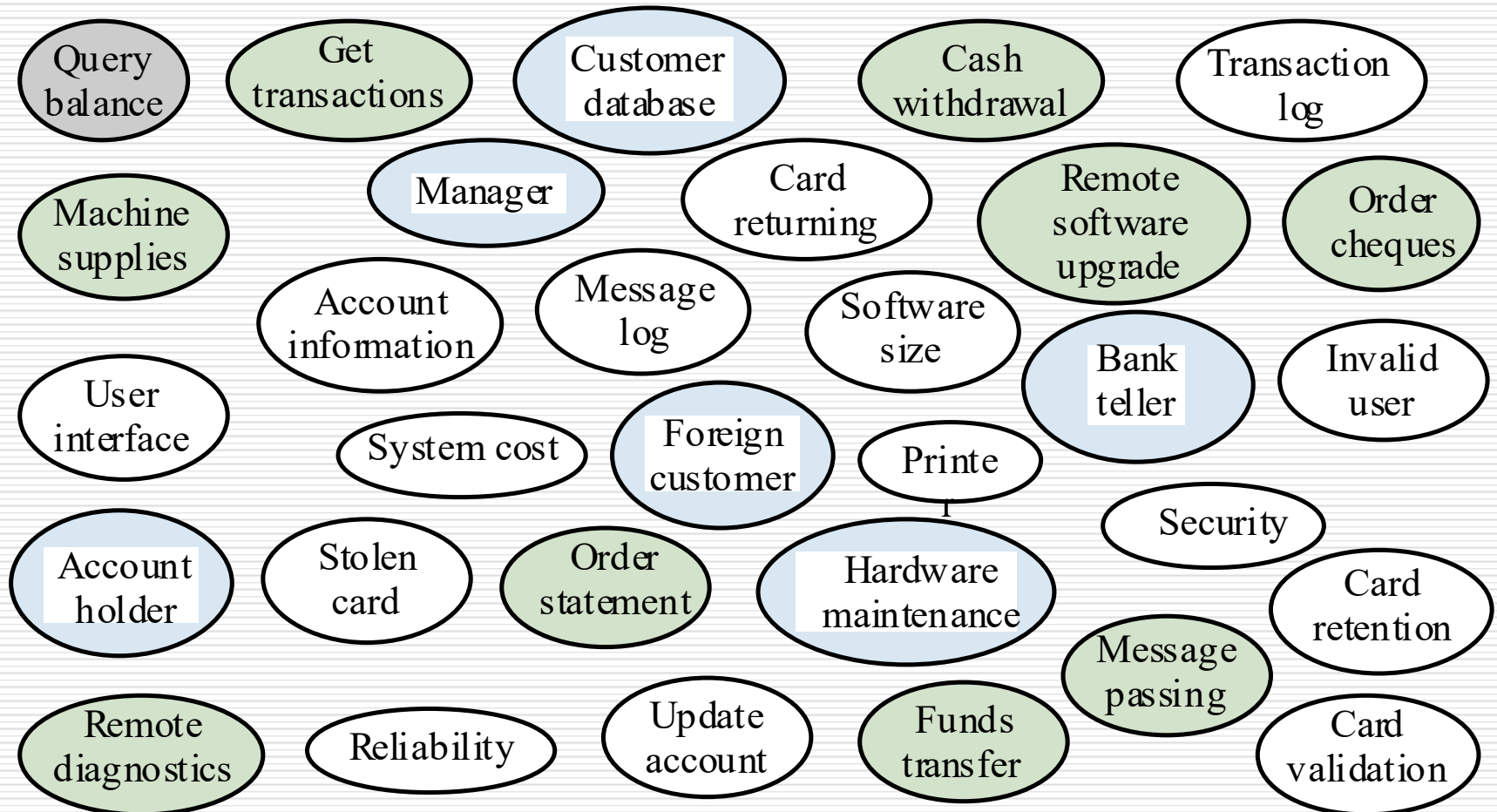
Viewpoint identification

- **Identify viewpoints using**
 - ✓ **Providers and receivers of system services;**
 - ✓ **Systems that interact directly with the system being specified;**
 - ✓ **Regulations and standards;**
 - ✓ **Sources of business and non-functional requirements.**
 - ✓ **Engineers who have to develop and maintain the system;**
 - ✓ **Marketing and other business viewpoints.**

An example: ATM viewpoints

- ✓ **Bank customers**
- ✓ **Representatives of other banks**
- ✓ **Hardware and software maintenance engineers**
- ✓ **Marketing department**
- ✓ **Bank managers and counter staff**
- ✓ **Database administrators and security staff**
- ✓ **Communications engineers**
- ✓ **Personnel department**
- ✓ **.....**

Viewpoint identification



(4) Videotape camera

- ✓ set up videotape cameras within workplace to record exactly what is being do.**
- ✓ play back the tape to obtain requirement information.**
- ✓ 仿真，仿制，模仿，学习引进先进系统**

xxxx年秋季入学新生报到、注册现场部分图片





学生先领取录取通知书、学籍卡片、学生证



学生在老师指导下填写学生证、卡片



学生交费后统一换取正式发票



学生办完手续后，买取教材



学生凭发票交回卡片，领取光盘、学生手册、盖章

(5) questionnaire, inquire table

(调查表, 问讯表)

某出版社系统调查表

编号	提出问题
1	您在哪个部门工作？
2	出版业务流程是什么？
3	您每日都处理那些文件、数据、报表？
4	工作中手工处理特别麻烦的事情是什么？
5	工作中手工处理什么问题解决不了？影响效率的问题有哪些？
6	您认为提高工作效率，节省工作时间，减轻工作强度可采取哪些办法？

某出版社系统调查表

编号	提出问题
7	您的部门需要成本核算和统计的内容有哪些？
8	您的部门采用计算机管理工作情况如何？
9	如何改进业务流程使之更合理？
10	哪些问题是目前传统手工方法根本无法解决的？
11	出版社计算机管理信息系统需要解决什么问题？

(6) Web searching

➤ 互联网是最聪明的老师

- ✓ 海量的信息库
- ✓ 人类的知识库
- ✓ 万能的Web
- ✓ 向互联网学习，找答案
- ✓

(7) 网上需求社区

- 正如开源社区一样
- 通过开放讨论，共同构思需求，共享提出的需求
- 例如：
 - ✓ 未来手机的需求
 - ✓ 智能衣服的需求
 - ✓ 智能家居的需求

Requirements Gathering(技术比较)

➤ Interviews

- Same Place, Same Time
- Few People, Analyst-Driven

➤ Questionnaires

- Different Time, Different Place
- Many People, Analyst-Observer

➤ Group Sessions

- Same or Different Place, Same Time
- <20 People, Analyst-Facilitated

➤ Observation

- Same Time, Same Place
- Analyst-Observer

- Documentation Review
- Brainstorming
- Apprenticing
- Scenario Analysis
- Prototyping/Mock-up
- Modeling
- Workshops

A user requirement must be ...

- ✓ **documented**
- ✓ **expressed precisely**
- ✓ **expressed as what, not how**
- ✓ **prioritized**
 - **essential, desirable, optional**
 - **primary, secondary, tertiary**
- ✓ **testable**
- ✓ **covered in the design**
- ✓ **implemented**
- ✓ **traceable (in order to assure the last 3)**

什么是高质量的用户需求？

- **真实性**：真实地反映利益相关方的需求
- **一致性**：需求项内部、需求项之间没有逻辑冲突
- **精确性**：需求的表述不会引起二义或者多义理解
- **无冗余**：每项需求在软件需求模型中仅出现一次，多项需求之间不存在语义重叠
- **完全性**：所有需求项构成的全集完整地覆盖所有利益相关方的需求，尤其不能遗漏重要或紧迫的需求
- **可行性**：在实际资源约束条件下，需求能够被完整实现的可能性
- **可验证性**：在验收测试阶段，通过呈现测试结果，能够客观地、无争议地向利益相关方表明需求已完整实现

发现软件需求描述的缺陷

火车自动售票机控制软件的需求描述：

- ❑ 当用户点击“开始”按钮，包含所有潜在目的地的菜单被激活。
- ❑ 在用户选择目的地和乘车日期之后，系统要求用户插入银行卡。系统检查银行卡的有效性，并要求用户插入身份证。
- ❑ 在验证身份证的有效性之后，系统从信用卡扣款并快速弹出车票。

在真实性方面是否存在缺陷？

□ 当用户点击“开始”按钮，包含所有潜在目的地的菜单被激活

■ 用户并不关心是否必须通过一个具体的“开始”按钮来启动购票流程

□ 在用户选择目的地和日期后，系统要求用户插入银行卡。系统检查银行卡的有效性，并要求用户插入身份证

■ 购票时只需提供身份证号即可

一致性？

- 在用户选择目的地和日期后，系统要求用户插入**银行卡**。
系统检查**银行卡**的有效性，并要求用户插入身份证。
 - 在验证身份证的有效性之后，系统从**信用卡**扣款并快速弹出车票。
- 除信用卡之外，是否需要支持具有支付功能的其他银行卡？

精确性？

- 当用户点击“开始”按钮，包含所有潜在目的地的菜单被激活。

■ 同城多站

- 在用户选择目的地和日期后，系统要求用户插入银行卡。系统检查银行卡的有效性，并要求用户插入身份证。

■ 银行卡的有效性有两种含义：(1) 未被冻结，可正常使用；(2) 除前一含义外，还要求余额足够支付票款

- 在验证身份证的有效性之后，系统从信用卡扣款并快速弹出车票。

完全性？

- 当用户点击“开始”按钮，包含所有潜在目的地的菜单被激活。
- 在用户选择目的地和日期之后，系统要求用户插入银行卡.....
 - 必须支持用户手工输入目的地，并且系统动态地根据用户输入筛选潜在的目的地
 - 如果存在多条乘车路线，怎么办？

可行性？

□ 当用户点击“开始”按钮，包含所有潜在目的地的菜单被激活。

■ 全国所有潜在的目的地数量过于庞大！

可验证性？

- 在验证身份证的有效性之后，系统从信用卡扣款并快速弹出车票。
- 不能客观地验证是否“快速”

改进的需求描述

- ❑ 用户输入目的车站的名字，系统动态地显示匹配的目的地车站的清单。
- ❑ 在用户从清单中选定目的地车站和乘车日期之后，系统给出所有可能的乘车路线及每条路线的车票价格。
- ❑ 用户从中选定一条路线之后，系统显示车票价格并提示用户插入银行卡、输入身份证号。系统也可以接收来自身份证识别装置传入的身份证号。
- ❑ 系统检查银行卡及身份证号的有效性，检查银行卡余额是否足够支付票款。检查通过后，系统从银行卡扣款并在用户输入齐备后2秒之内弹出车票。

Initial requirements

An example

Case Study: SafeHome

Product description:

Our research indicates that the market for home security systems is growing at a rate of 40% per year. We would like to enter this market by building a microprocessor-based home security system that would protect against and/or recognize a variety of undesirable “situations” such as illegal entry, fire, flooding, and others. The product, tentatively called *SafeHome*, will use appropriate sensors to detect each situation, can be programmed by the homeowner, and will automatically telephone a monitoring agency when a situation is detected. ...

Case Study: SafeHome (Cont.)

Objects

- smoke detectors, window and door detectors, motion detectors, an alarm, an event, a control panel a display, telephone numbers, a telephone call, ...

Services

- setting the alarm, monitoring the sensors, dialing the phone, programming the control panel, reading the display

Constraints

- manufactured cost of less than \$80, user-friendly, interfacing with a standard phone line

Performance Criteria

- a sensor event recognized within one second

Case Study: SafeHome (Cont.)

Example of **mini-specifications** (object: control panel)

- mounted on wall
- size approximately 9 X 5 inches
- contains standard 12-key pad and special keys
- contains LCD display of the form shown in sketch
- all customer interaction occurs through keys
- used to enable and disable the system
- software provides interaction guidance, echoes, and the like
- connected to all sensors

原始需求难理解，正确性不能保证

原始需求与软件设计有什么关系？

Requirements validation techniques

☐ Requirements auditing

- ✓ Systematic manual analysis of the requirements.

☐ Prototyping

- ✓ Using an executable model of the system to check requirements.

☐ Test-case generation

- ✓ Developing tests for requirements to check testability.

☐ Modelling.....

3.5 Model and Modeling

- ✓ Once an initial requirements has been determined, next step is to analysis and refine them (时机)
- ✓ Modeling help to analysis user requirements and change user requirement into **software requirement**.
- ✓ Model is also a part of specification from different perspectives and is a **bridge for design**.

模型和建模

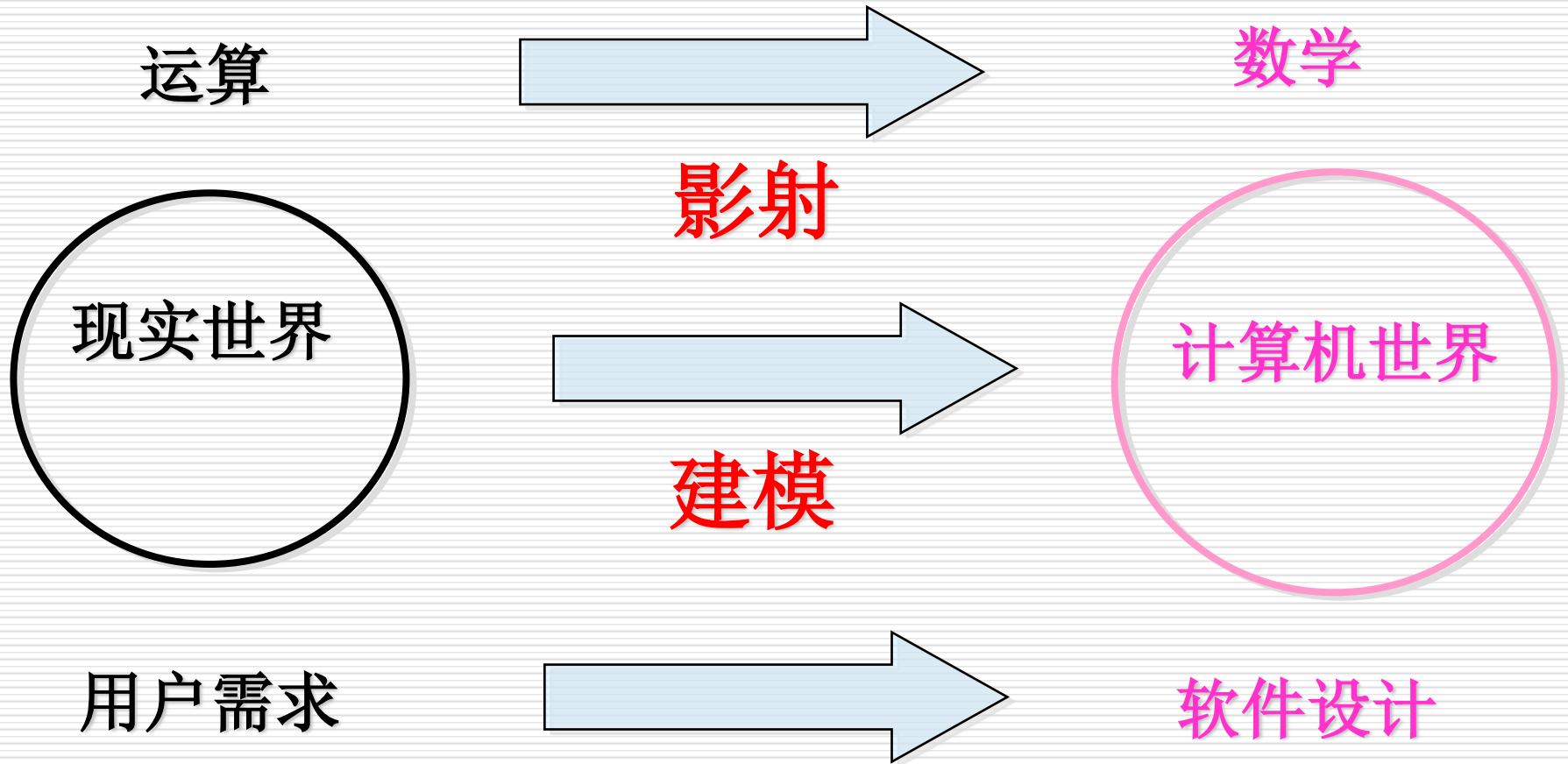
模型是现实世界某些重要方面的表示。

模型一种抽象，从某个视点、在某种抽象层次上详细说明被建模的系统。

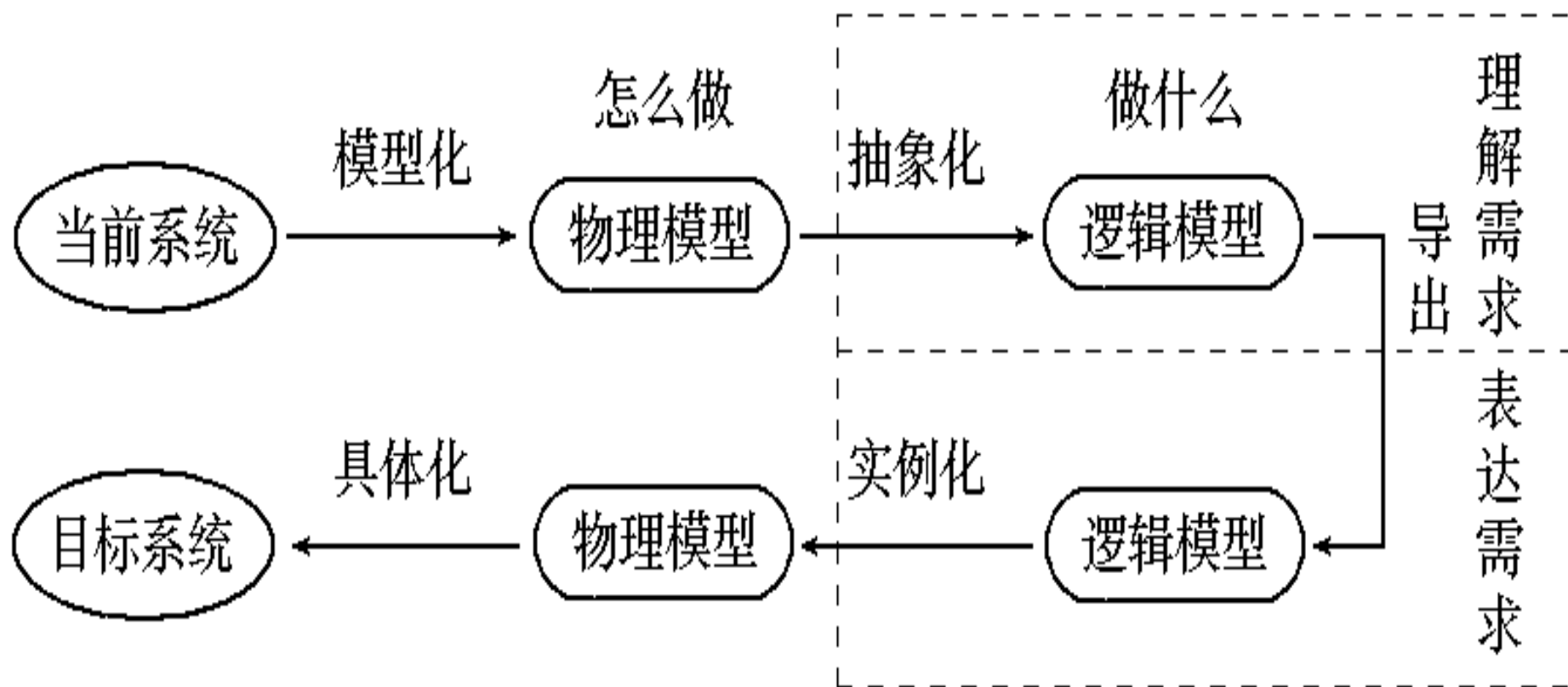
有时我们使用术语“**抽象**”来表示模型，因为我们从现实世界中抽象出对我们特别有用的东西。

模型一般分为具体模型和抽象模型两大类。具体模型有直观模型、物理模型等；抽象模型有思维模型、符号模型、图形模型、描述模型、数学模型等。

How to get model? 建模, 抽象化.....



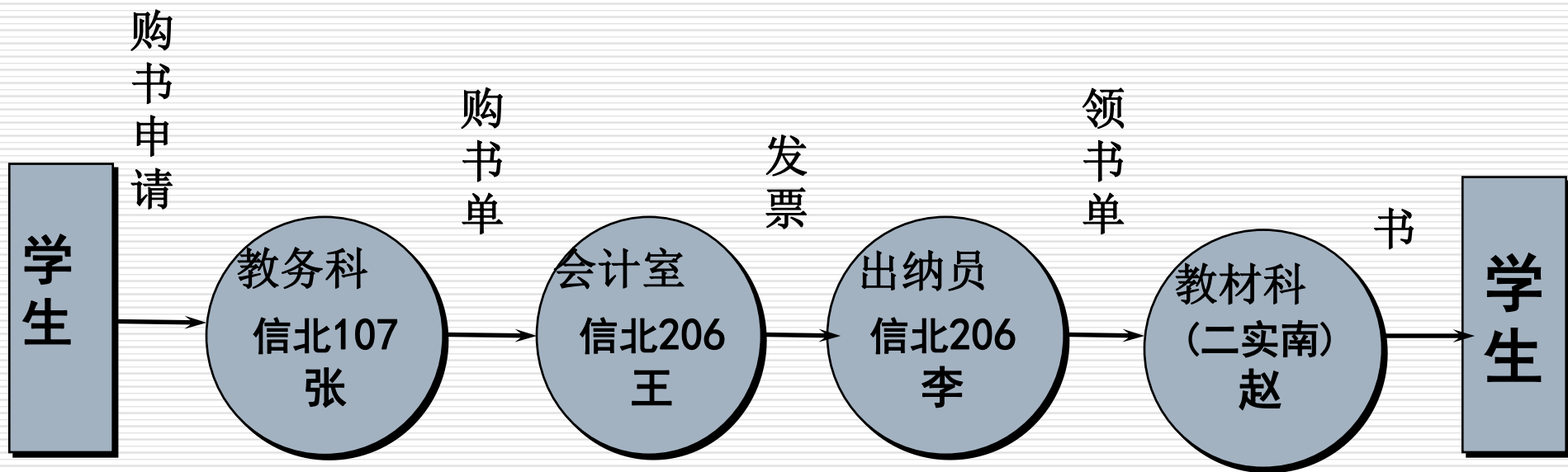
How to get model? 建模, 抽象化.....



建模分析过程 — 举例1

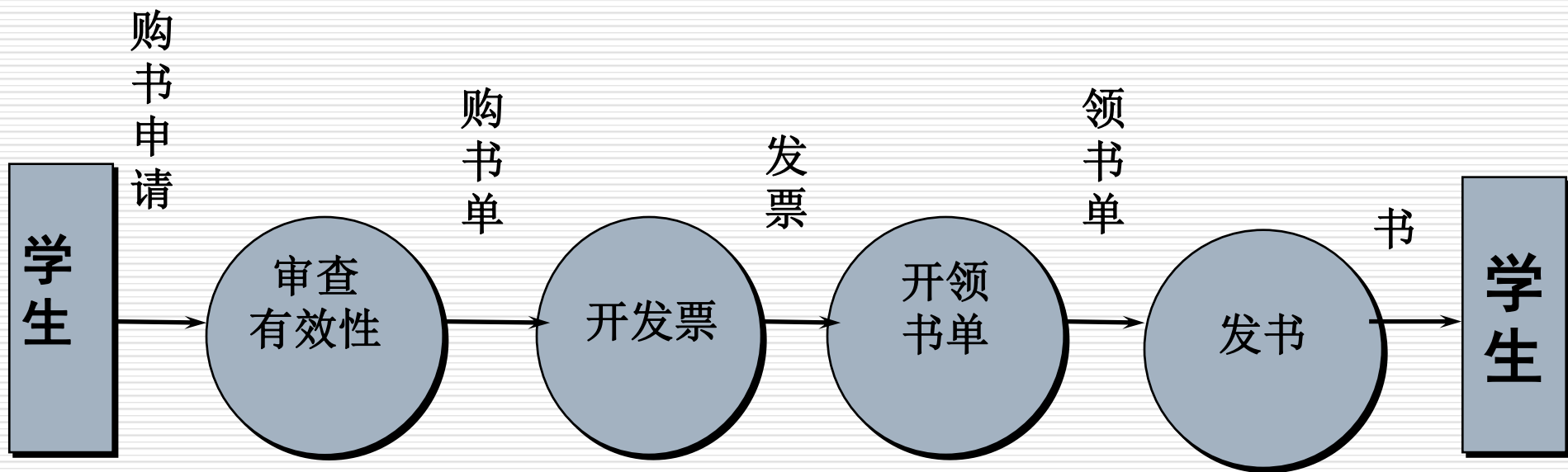
学生购买教材的实际处理流程 — 当前系统物理模型

(1) 通过对现实环境的调查，获得当前系统的物理模型



建模分析的过程

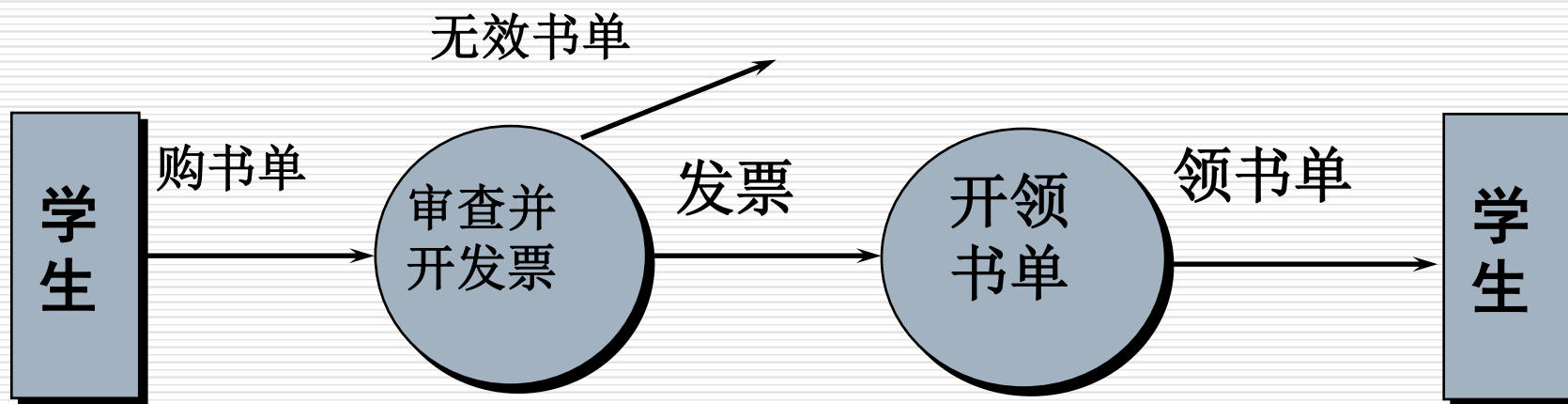
(2) 去掉具体模型中的非本质因素，抽取现实系统的实质，抽象出当前系统的逻辑模型。



学生购买教材的逻辑模型

建模分析的过程

(3) 分析当前系统与目标系统的差别，建立目标系统的逻辑模型



计算机教材管理系统的逻辑模型

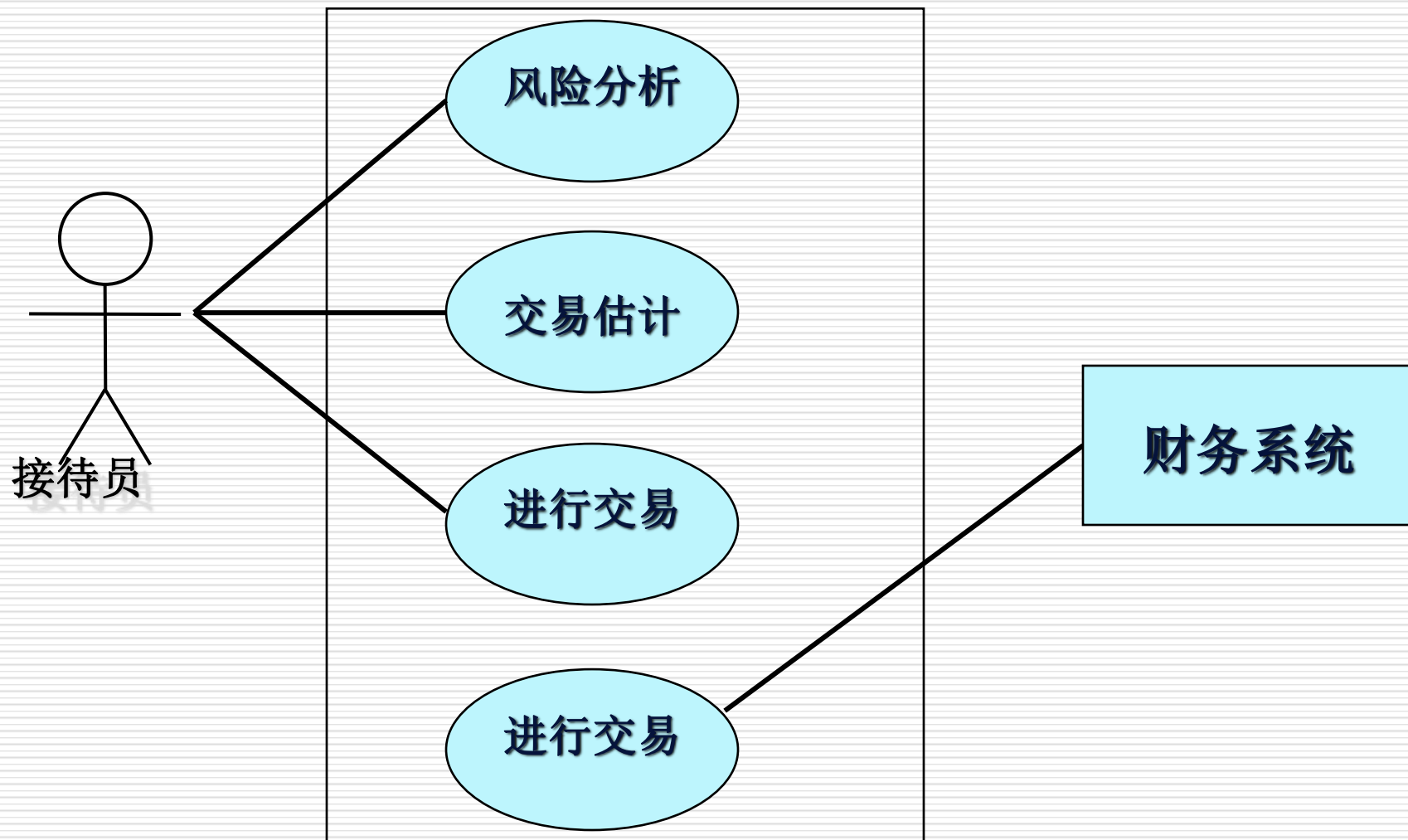
建模分析过程示意

(4) 对目标系统的逻辑模型进行改进与优化

(5) 需求分析的验证

建模分析过程 — 举例2

某金融贸易系统--用例图



总结：模型的作用

- 在建模过程中了解系统
- 通过抽象降低复杂性
- 有助于回忆所有的细节
- 有助于开发小组间的交流
- 有助于与用户的交流
- 为系统的维护提供文档

模型的作用

- ✓ **模型化或模型方法是通过抽象、概括和一般化，把研究的对象或问题转化为本质（关系或结构）相同的另一对象或问题，从而加以解决的方法。**
- ✓ **模型化方法要求所建立的模型能真实反映所研究对象的整体结构、关系或某一过程、某一局部、某一侧面的本质特征和变化规律。**

软件需求分析模型

- 数据流图（DFD）
- 实体—联系图（ **ERD** ）
- 类图
- 实例图
- 时序图
- 状态图
- 协作图
- 事件列表
- 数据字典模型
- 数据元素定义
-

§ 3.6 软件需求分析建模方法

- 结构化分析(传统建模方法)
- 面向对象分析